

Otto Ninja Starter robot

Otto Robot

VIEW IN BROWSER

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Summary

Unique open source robot that walks and transforms into a wheeled racer! made by Sebastian Coddington.

11.12 hrs	4 pcs	0.20 mm	0.40 mm	PLA	126 g	Prusa MINI / MINI+

Hobby & Makers > RC & Robotics

Tags: robot stem arduino esp8266 esp32 ninja ottodiy edtech

Build your own robot like a Ninja 🐱

This is the first emotional modular EdTech Arduino opensource robot that can walks & rolls!

Features

- **Walk & dance**
- **Transforms quickly into a wheeled robot**
- Simple Programming with Otto Blockly or Arduino

- Expand to Humanoid here: Otto Ninja Humanoid 3D files
- Avoid obstacles with ultrasonic sensor
- LED matrix or OLED emotional eyes to swap
- Makes emotional sounds and melodies
- Button for interactions
- **Wireless communication – for remote controlling**

Print instructions

Otto is very well-designed for 3D printing, so won't give you trouble if you follow this common parameters:

Recommended using an FDM 3D printer.
No need of supports or rafts.
Resolution: 0.2 mm or less
Fill density 15%

List of Parts

- Otto Ninja PCB ESP8266 (you could use a wemos board with a servo shield)
- Micro USB cable Data Sync
- Battery 6F22 Rechargeable Lithium (9V 650mAh with USB port for charging)
- Toggle switch + UBEC Regulator (OUT 5V/ 3A) + XH 2P connector + 9V Battery connector soldered.
- 2 x Ninja Servo 180° metal gears (it comes with 3 screws)
- 2 x Ninja Servo 360° continuous rotation metal gears (it comes with 3 screws)
- 2 x Oring 68 mm OD 60 mm ID 4 mm
- Ultrasonic sensor HC-SR04
- Button module with headers
- 4pin Dupont cable with connector
- 2 x 3pin Dupont cable with connector
- Buzzer module with headers
- Screwdriver Phillips 2.5x40mm (with magnetic tip)
- 4 x Metal self-tapping screw M2*5 (they must be ferromagnetic)
- Matrix 16x8 LEDs HT16K33 (**OPTIONAL)**
- OLED display 1.3" (**OPTIONAL)**
- 3D printed Ankle Left Black
- 3D printed Ankle Right Black
- 3D printed Base Black
- 2 x 3D printed Foot Black
- 3D printed Inner Bottom Black
- 3D printed Legs Black
- 3D printed Starter Lid Black

- 3D printed Plate Matrix Lid Black
- 3D printed Plate Ultrasonic Lid Pink
- 3D printed Band Red

Buy similar kit here HP-Otto Invent Expansion - Creator KitHow to Build videoWhat can Otto Ninja do?

Otto Ninja can walk, transform, roll, show emotions, dance, sense motion & objects

Coding from beginner to advanced

Program Otto Ninja to think on its own using block-based coding with Blockly or C++ with Arduino IDE.

Github repo here OttoDIY/OttoNinja

Want to just play?

Via the kit's **wireless communication** module, remote control your Otto Ninja by connecting to your mobile or tablet to race or have fun battles!

Alternatively you can try to build the previous design that uses more common off the shelf parts Otto Ninja robot Arduino Nano design by Otto DIY | Download free STL model | Printables.com

We invested lots of time and resources to provide open source code, software and hardware, please support this project by just **giving us a ❤ Like, and share**

You are **welcome to be a part of this friendly community** of robot builders, teachers and makers.

Join today our Otto Builder community

This remix is based on



Otto DIY build your own robot

by Otto Robot



Otto DIY Wheels robot

by Otto Robot



Otto Ninja robot Arduino Nano design

by Otto Robot



Otto DIY Smart OLED robot

by Otto Robot



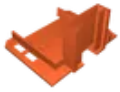
Otto DIY Emotional LED eyes robot

by Otto Robot

Model files



otto-diy_ninja_lid.stl



otto-diy_ninja_innerplate.stl



otto-diy_ninja_head.stl



otto-diy_ninja_legs.stl



otto-diy_ninja_leg-right.stl



otto-diy_ninja_leg-left.stl

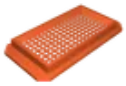


otto-diy_ninja_footwheels-x2.stl

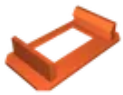
3D print 2



otto-diy_ninja_plate-ultrasonic.stl



otto-diy_ninja_plate-matrix.stl



otto-diy_ninja_plate-oled.stl

OPTIONAL



otto-diy_ninja_band.stl

print in red :D

otto-ninja-2021-2-v14.step

CAD file

Print files



otto-diy_ninja_mainbodyparts_02mm_pla_mini_8h46.gcode

PLA 0.40 mm 0.20 mm 8.76 hrs 96 g Prusa MINI / MINI+

Main Body parts in black



otto-diy_ninja_foot-x2_02mm_pla_mini_1h12m.gcode

PLA 0.40 mm 0.20 mm 1.20 hrs 18 g Prusa MINI / MINI+

Feet/ Wheels in black



otto-diy_ninja_plates_02mm_pla_mini_41m.gcode

PLA 0.40 mm 0.20 mm 0.69 hrs 8 g Prusa MINI / MINI+

Modular eyes plates in Pink



otto-diy_ninja_band_02mm_pla_mini_28m.gcode

PLA 0.40 mm 0.20 mm 0.47 hrs 4 g Prusa MINI / MINI+

Ninja band in red

Find source .stl files on Thingiverse.com

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